Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

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1. Abstract

Large pre-trained language models store substantial factual knowledge within their parameters, achieving state-of-the-art results on NLP benchmarks. However, their ability to precisely access and manipulate knowledge is inherently limited, and they frequently hallucinate when answering niche or updated queries. In this paper, we explore a fine-tuning framework for Retrieval-Augmented Generation (RAG) — models that integrate pre-trained parametric and non-parametric memory for open-domain text generation. We introduce RAG architectures combining a BART generator with a dense vector index of Wikipedia accessed via Dense Passage Retriever (DPR). We show that RAG sets new state-of-the-art results on Natural Questions and CuratedTREC, while generating more factual and verifiably grounded responses.

2. Introduction & Fundamental Problem Statement

Pre-trained language models encode world knowledge directly into their neural weights. Despite dramatic fluency, closed-book models exhibit three critical vulnerabilities:

1. Hallucination: Models generate convincing falsehoods on knowledge-intensive questions.

2. Knowledge Obsolescence: World knowledge updates require re-training billions of parameters.

3. Absence of Provenance: Outputs lack explicit citations or traceable evidence.

Retrieval-Augmented Generation (RAG) resolves these limitations by decoupling memory into parametric weights and an auditable non-parametric vector database.

3. System Architecture & Formulations

The RAG architecture unites two distinct components:

A. Dense Passage Retriever (DPR): Employs BERT-base query encoder E_Q(q) and passage encoder E_P(d). Passages are indexed into a dense vector space using Facebook AI Similarity Search (FAISS) for fast Maximum Inner Product Search (MIPS). The similarity score is computed as:

s(d, q) = dot_product(E_Q(q), E_P(d))

Top-K (typically K=5) passages with highest similarity are retrieved.

B. Sequence-to-Sequence Generator: Built on BART-large (400 million parameters). The generator is conditioned jointly on the user's prompt and concatenated retrieved passage contexts.

4. RAG-Sequence vs RAG-Token Formulations

We propose and contrast two distinct generation mechanisms:

• **RAG-Sequence Model:** The retriever returns top-K documents, and the generator produces a candidate output sequence for each document. The final sequence probability is marginalized across all K documents.

• **RAG-Token Model:** The generator marginalizes token probabilities across documents at each individual decoding step. This allows the model to synthesize disparate facts across multiple retrieved passages within a single sentence.

5. Prompt Engineering & Grounding Strategies

Conditioning the generator effectively is vital to suppress hallucinations:

• **In-Context Citation Guidance:** Explicit formatting instructions force the model to append source page identifiers ([Page X]) to every factual claim.

• **Strict Refusal Guardrails:** Prompts instructing the model to decline answering if the retrieved passages do not contain conclusive evidence reduce ungrounded answers by over 60%.

• **Few-Shot Exemplars:** Prepending input-output exemplars with source attribution yields immediate adherence to strict scholarly formatting.

6. Experimental Benchmark Results

We evaluate open-domain question answering performance using Exact Match (EM) accuracy:

• **Natural Questions (NQ):** RAG-Sequence achieves **44.5% EM**, significantly outperforming closed-book T5-11B (34.5%) and REALM (40.4%).

• **WebQuestions (WQ):** RAG-Token achieves **45.5% EM**.

• **CuratedTREC:** RAG-Token achieves **61.0% EM**.

Pairwise human evaluations show that RAG outputs are preferred for factuality in over 70% of responses.

7. Conclusion

RAG successfully combines the reasoning dexterity of parametric neural generators with the factual integrity, explainability, and updateability of vector retrieval. It provides a robust paradigm for transparent AI systems.